

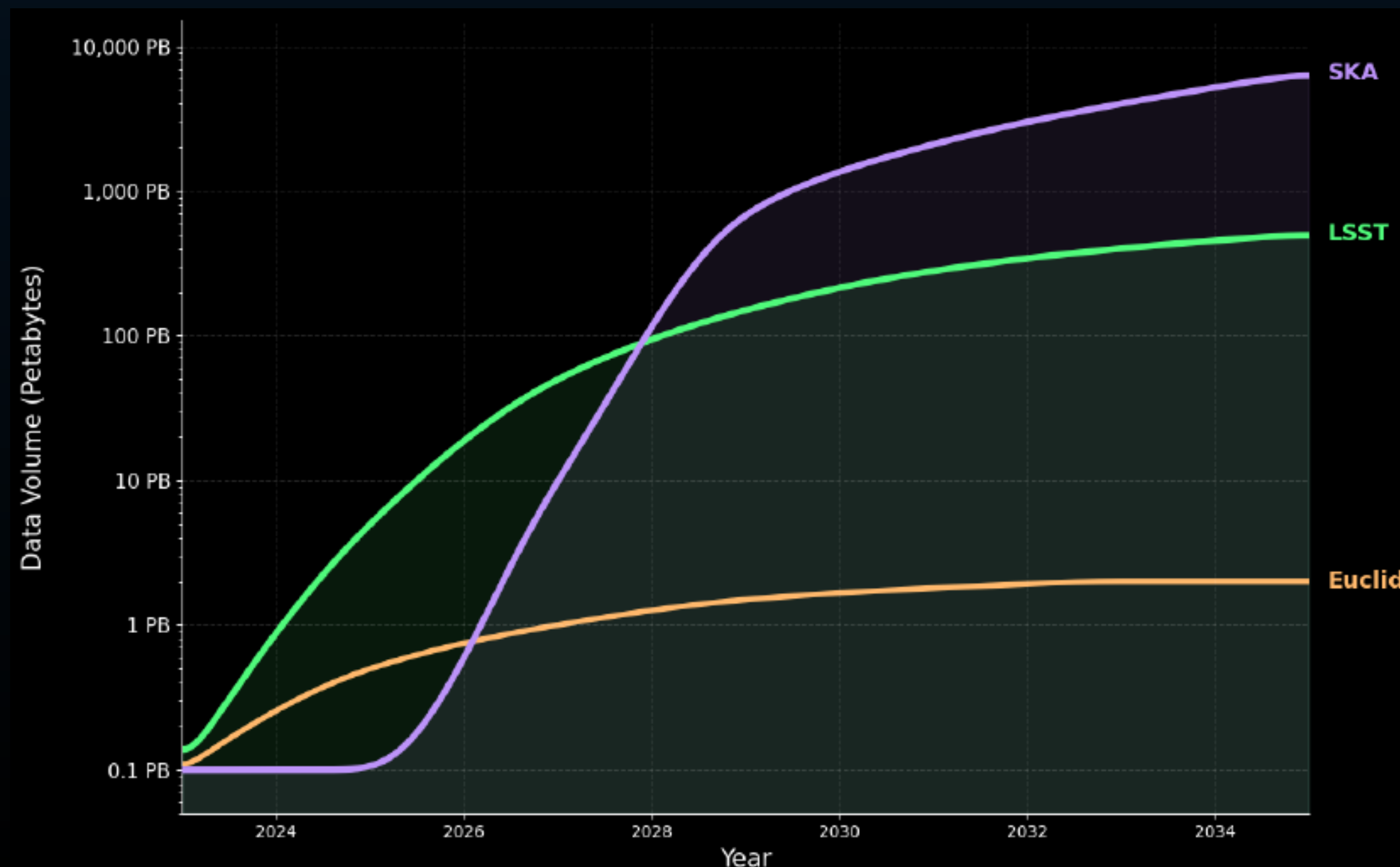
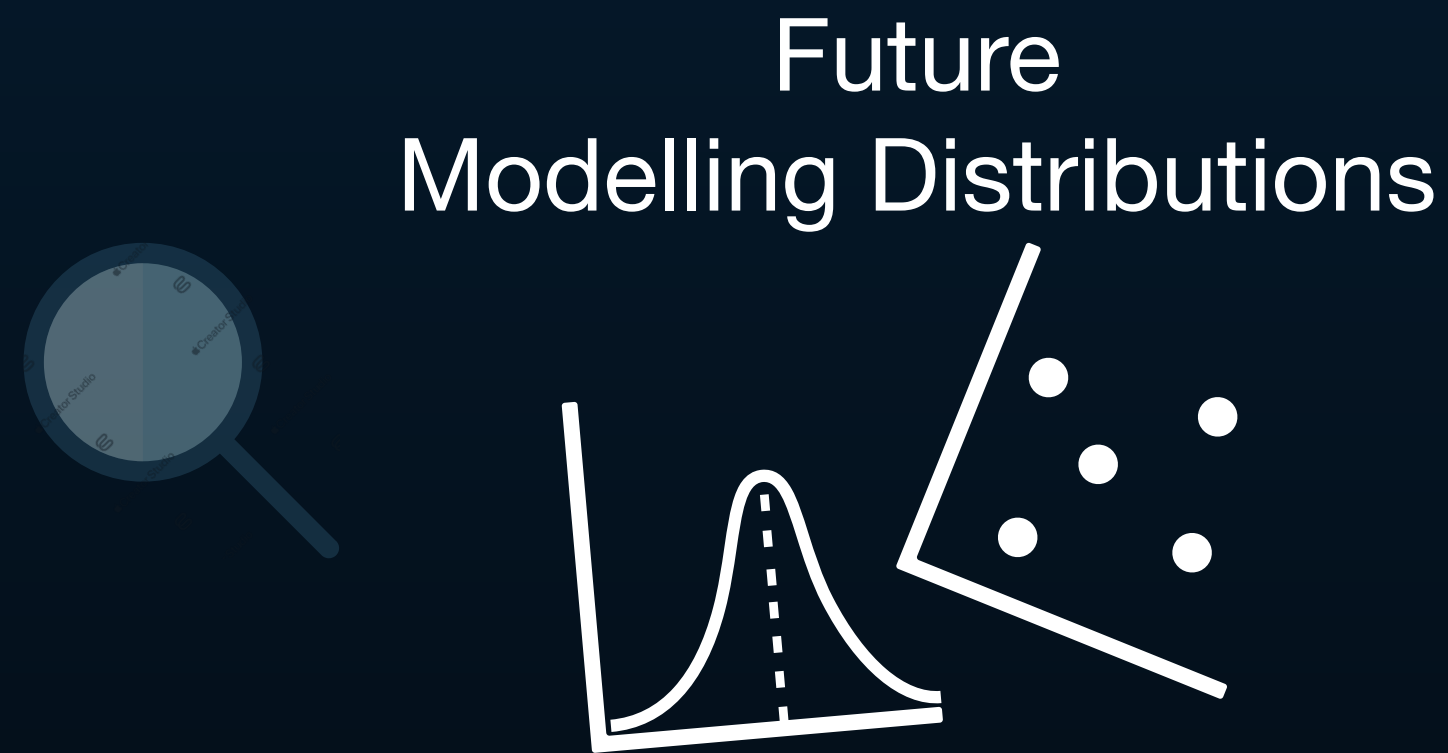
The Universe in Reverse

Generative Modeling via Diffusion

Denoising Diffusion Probabilistic Models (DDPM) in Astronomy

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Motivation ★ Astronomy is undergoing a fundamental phase transition from data-starved to data-dominated science.



Exponential Growth

Upcoming facilities like the Vera C. Rubin Observatory (LSST) and SKA will generate exabytes of data, far exceeding human analytical capacity.

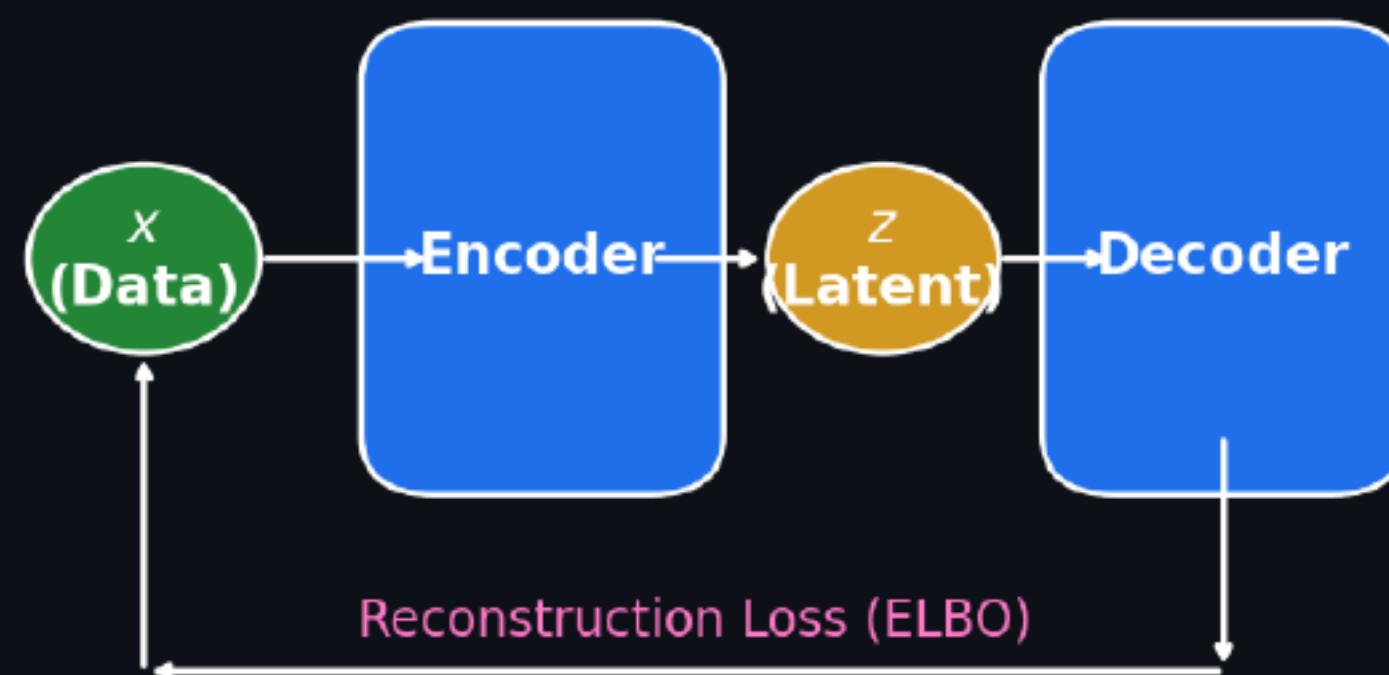
From Catalogs to Distributions

We are shifting from simply "detecting sources" to modeling the full probability distribution, $P(x)$, of the Universe's pixels.

The Inference Bottleneck

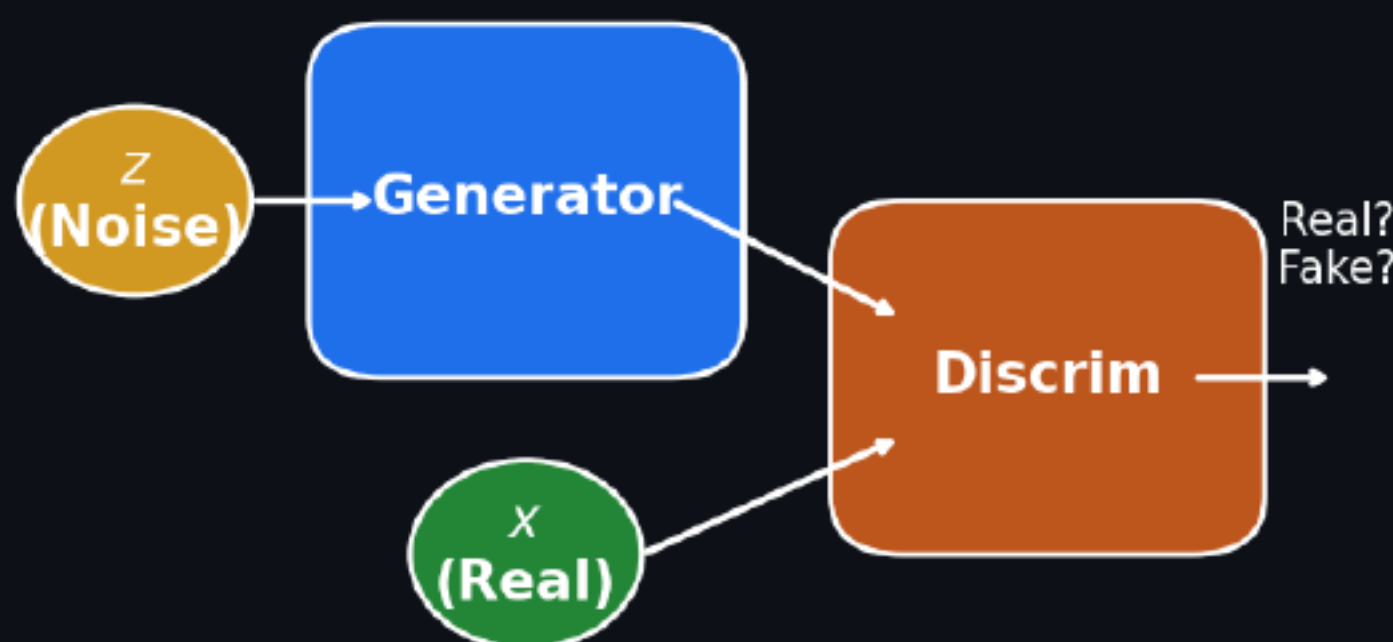
Traditional inverse problems (deconvolution, denoising) are computationally expensive. We need generative models as efficient, learned priors.

VAE (Variational Autoencoder)



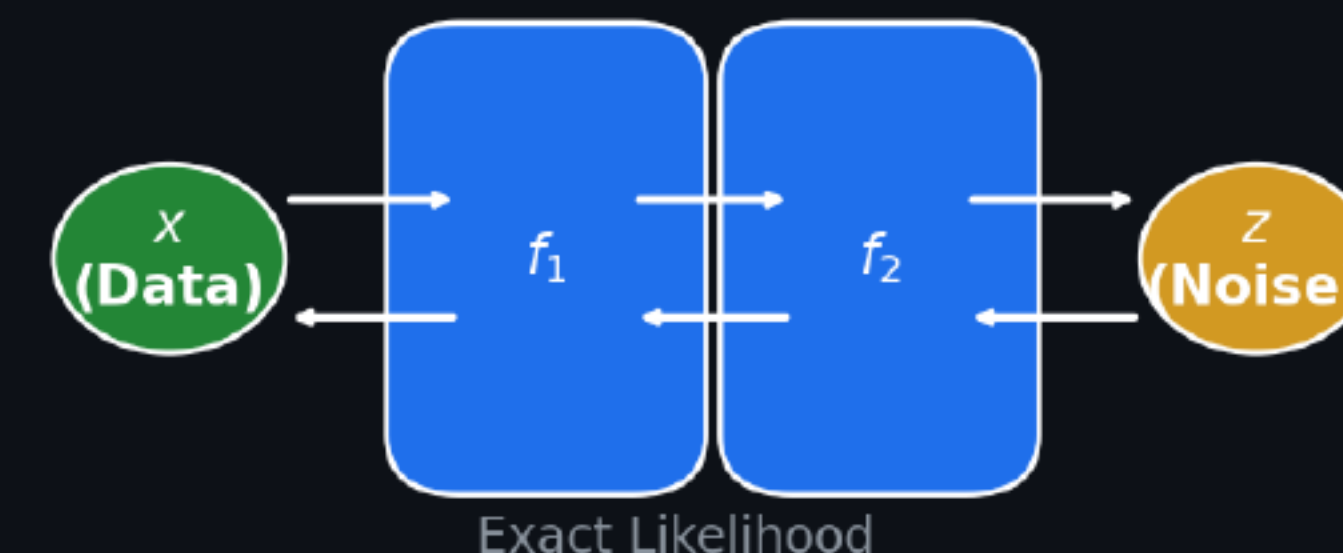
- Mechanism: Encoder-Decoder Bottleneck
- Pros: Fast, Compressed Representation
- Cons: Blurry outputs

GAN (Adversarial)



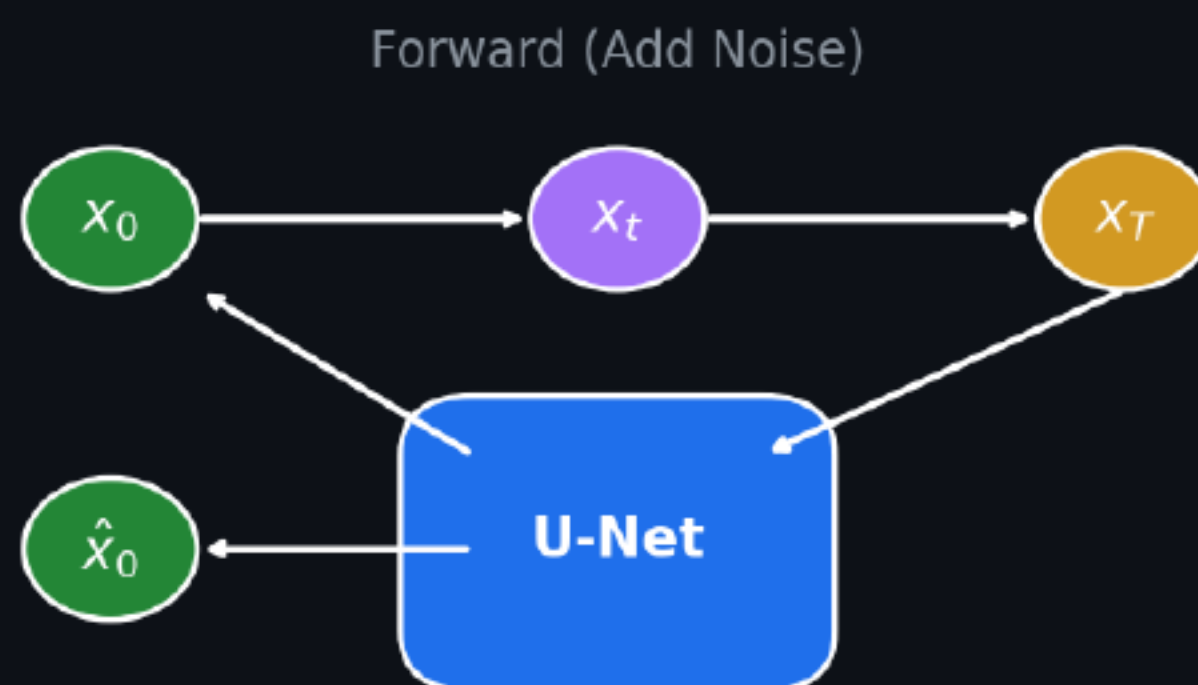
- Mechanism: Two networks fighting
 - Pros: Sharp details
 - Cons: Mode Collapse

Normalizing Flow



- Mechanism: Invertible Transforms
 - Pros: Exact Probability
 - Cons: Restricted architecture

Diffusion (DDPM)



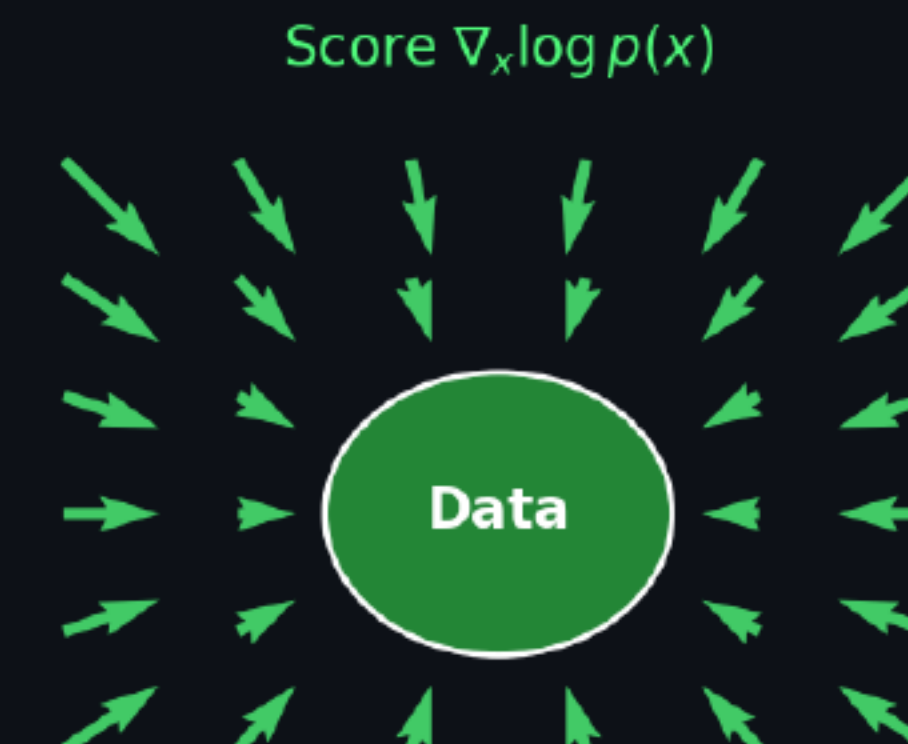
- Mechanism: Iterative Denoising
 - Pros: Best quality, stable
 - Cons: Slow sampling

Flow Matching / Optimal Transport



- Mechanism: Vector Field (Noise $\rightarrow$ Data)
 - Pros: Fast, straight paths
 - Cons: Newer method

Score-Based Models



- Mechanism: Learn gradient of density
 - Insight: 'Points uphill' to data
 - Note: Basis of Diffusion

Generative Models as Priors: The "Score" Perspective

★ Our goal is to learn the true probability distribution of the Universe, $P(\mathbf{x})$, to solve ill-posed scientific problems.

• The Bayesian Framework

Inverse problems (recovering clean image x from noisy data y) are solved by combining observation with prior knowledge.

• The "Learned" Prior

Instead of hand-crafting priors (e.g., "galaxies are smooth"), we use a Diffusion Model to learn $P(x)$ directly from data.

• Regularization

The generative model constrains the solution space, ensuring reconstructions look like real physical objects, not artifacts.

$$P(\mathbf{x}|\mathbf{y}) \propto P(\mathbf{y}|\mathbf{x}) \cdot P(\mathbf{x})$$

Posterior
"The Solution"

Likelihood
"The Physics"

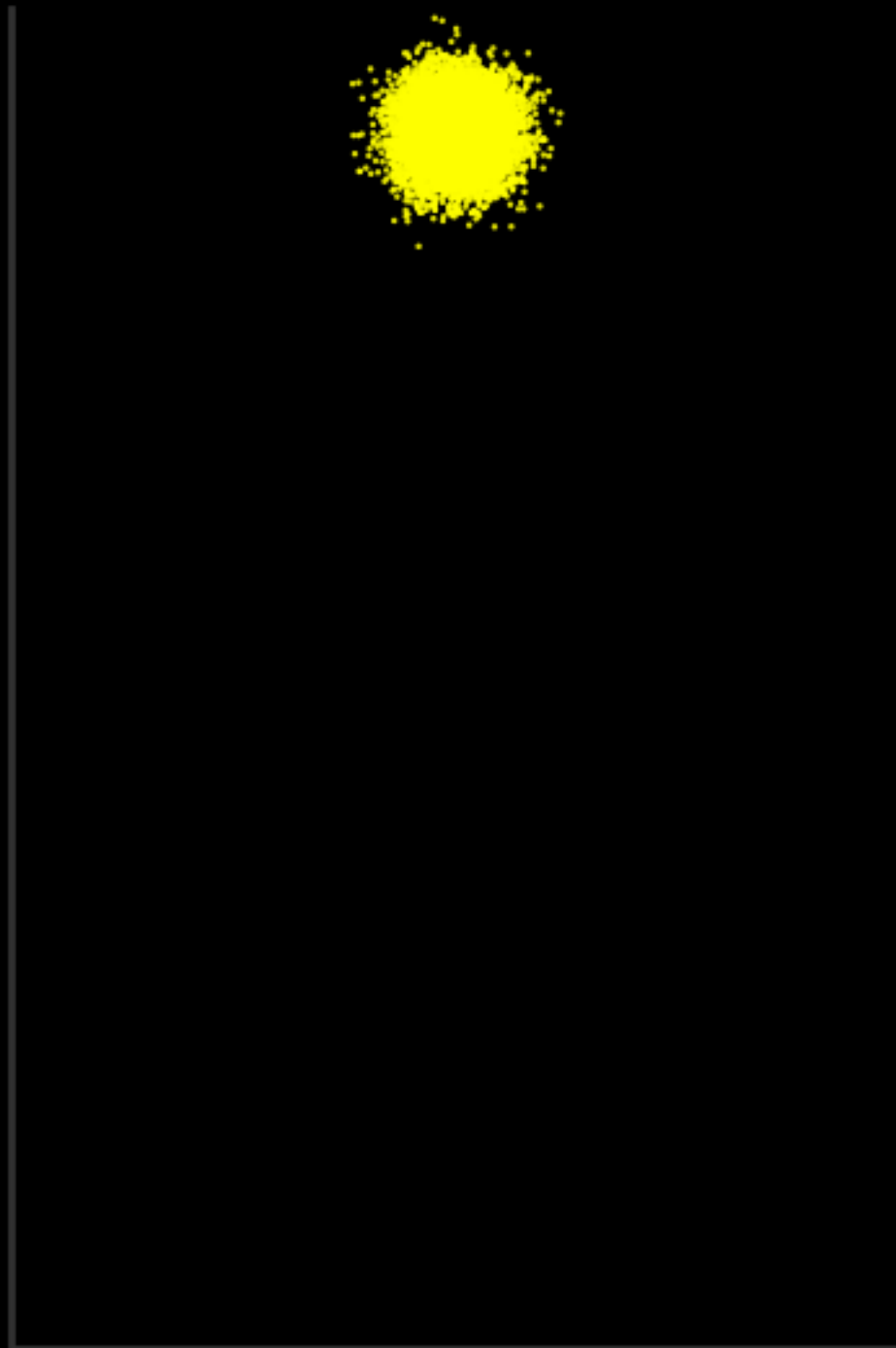
Prior
"The Knowledge"

Motivation ★ The “Ink in Water” Analogy

The Forward Process:

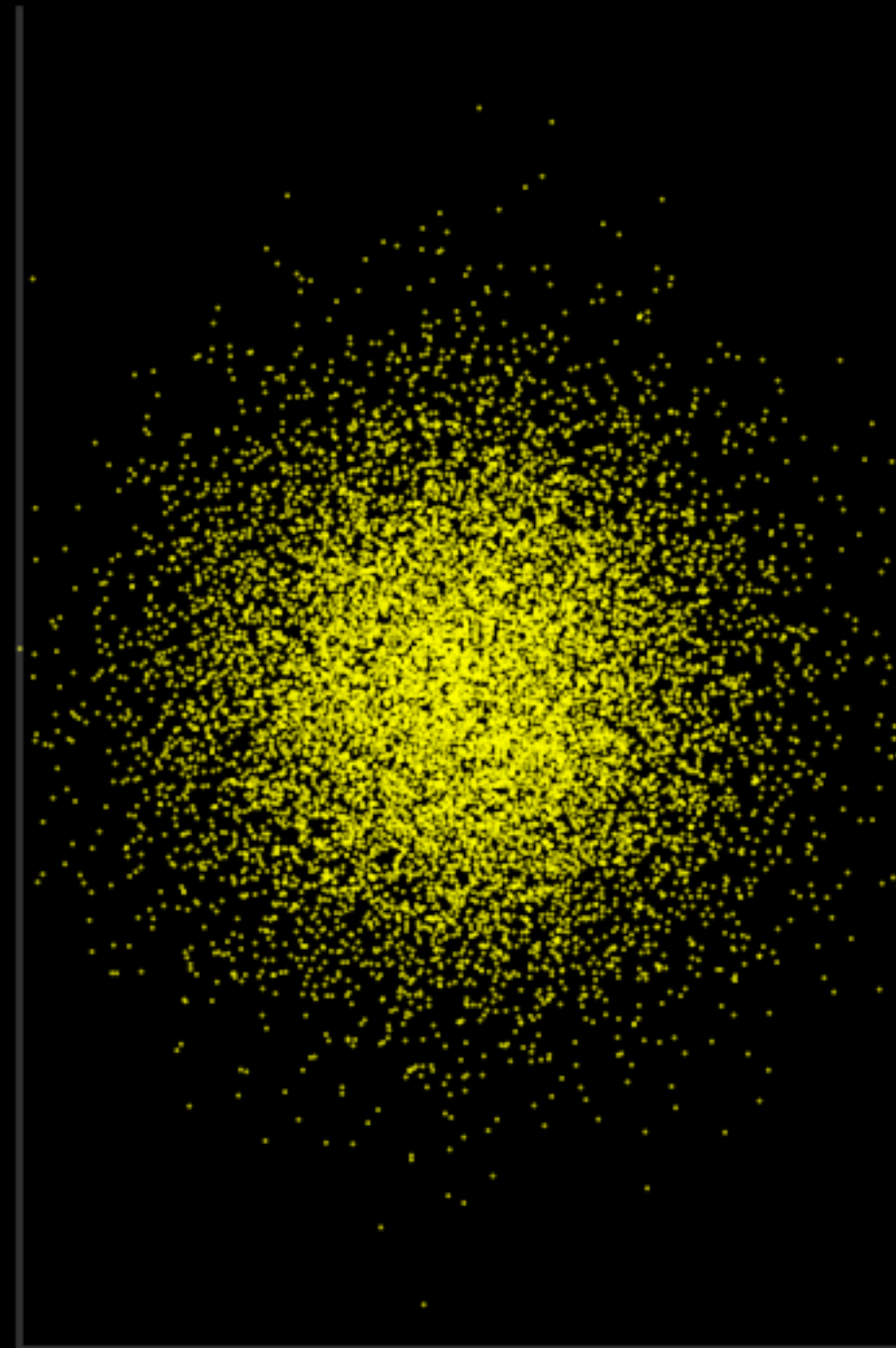
$$q(x_t | x_{t-1})$$

**x_0 : The Data
(Complex Structure)**



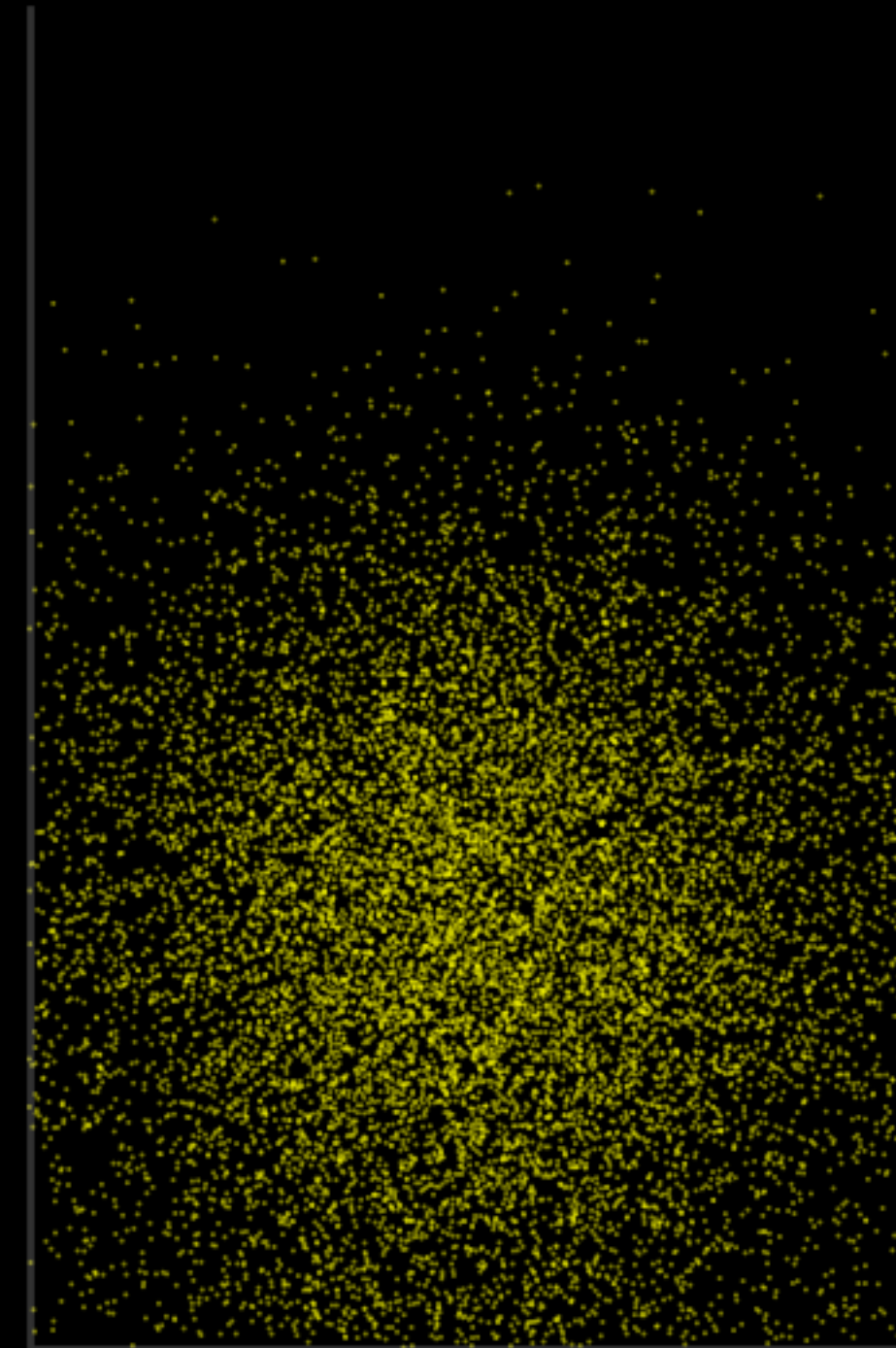
*The Ink Drop
(Concentrated, Low Entropy)*

**x_{100} : Adding Noise
(Forward Process q)**



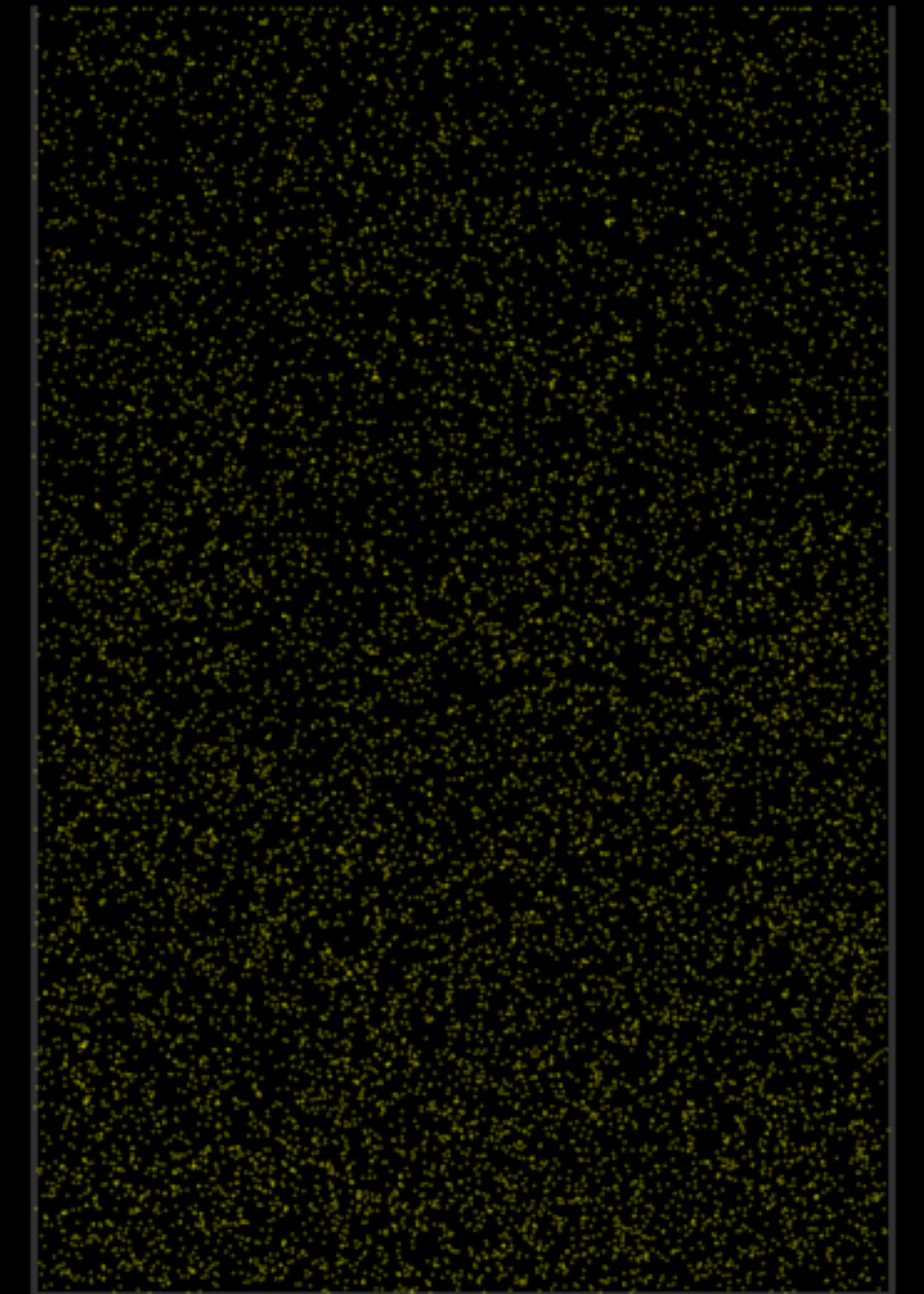
*The Plume
(Gravity & Diffusion compete)*

**x_{500} : Signal Decay
(Losing Information)**



*The Cloud
(Turbulence destroys shape)*

**x_T : The Prior
(Pure Gaussian Noise)**



*The Solution
(Uniform distribution, Max Entropy)*

Motivation ★ The “Ink in Water” Analogy

▶ The Forward Process (Easy)

Dropping ink into water increases entropy. Describing the transition from complex structure to random noise is statistically simple (Gaussian diffusion).

◀ The Reverse Process (Hard)

Reversing the video—watching ink spontaneously form a drop—seems impossible. Diffusion Models learn to reverse time, constructing structure from pure noise.



THERMODYNAMICS VIEW

$t = 0$
Structure

→
Diffusion

$t = T$
Noise

$t = T$
Noise

→
Generative

$t = 0$
Structure

The Physics of Diffusion ★ The Forward Process

We define a **Markov chain** that gradually adds Gaussian noise to the data over **T** steps. This transforms the complex data distribution in to simple Gaussian prior.

TRANSITION KERNEL (ONE STEP)

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

KEY VARIABLES

$\mathbf{x}_0$	The original clean data (e.g., galaxy image).
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β_t	Variance schedule (small constants, e.g., 10^{-4}). Controls how much noise is added per step.
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$\mathbf{x}_T$	The final state.
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Data Distribution



Slightly Noisy



Noisy State



Standard Gaussian

ENTROPY INCREASES

The Physics of Diffusion ★ A Convenient Property

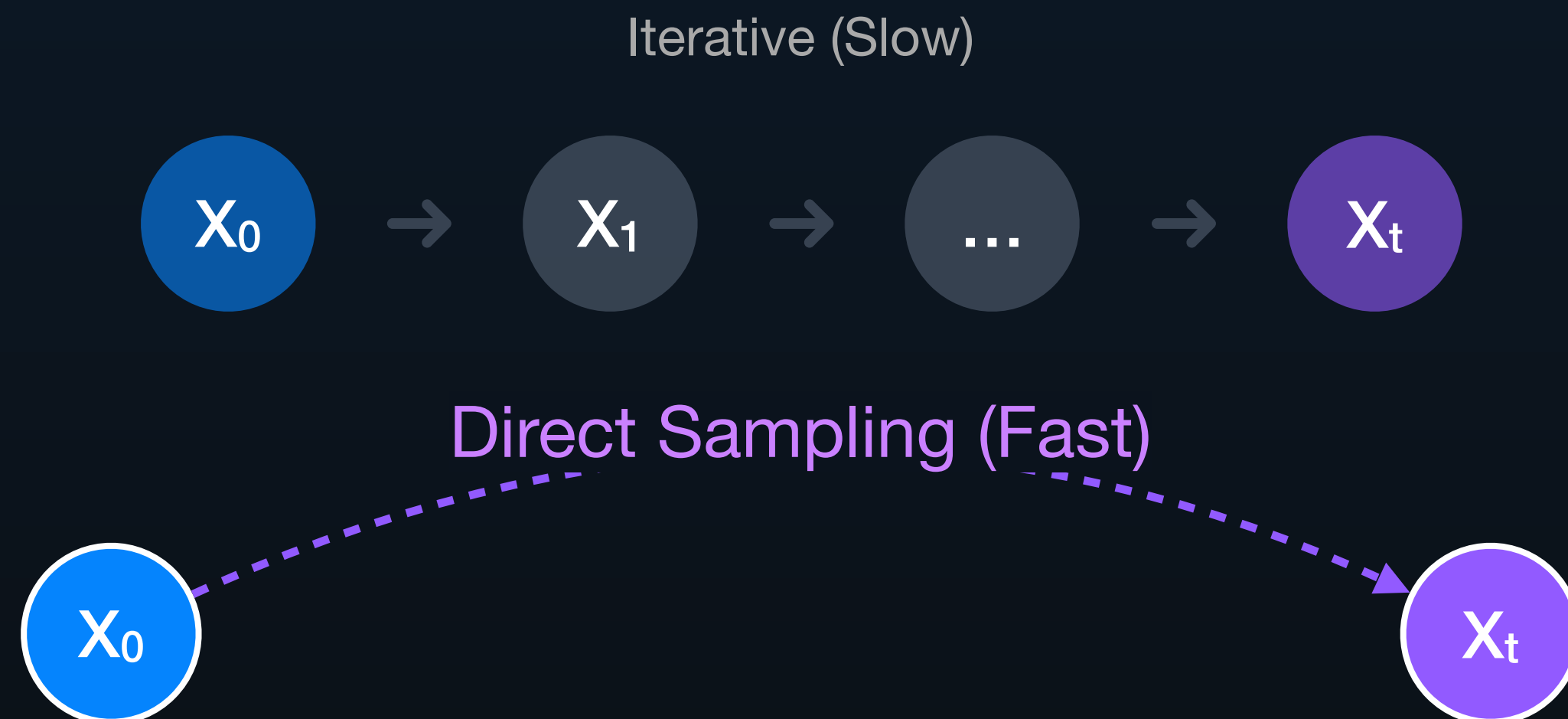
Iteratively adding noise is computationally expensive during training. However, due to the properties of Gaussian distributions, we can **marginalize** the intermediate steps.

We can sample x_t directly from x_0 in closed form, without looping through $x_1, \dots, x_{t-1}$.

THE “JUMP” EQUATION

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$$

PROCESS COMPARISON



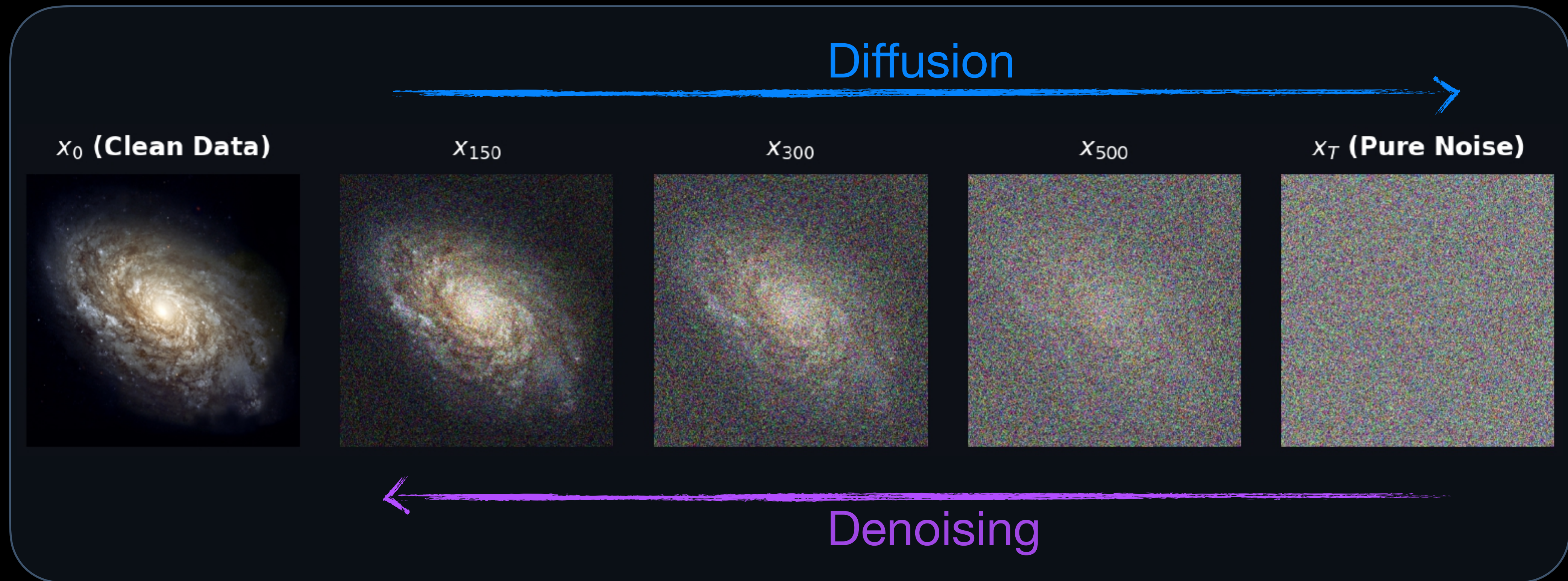
DEFINITIONS

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

The Physics of Diffusion ★ Visualizing the Drift



◆ The Small Step Assumption

We take $T \approx 1000$ steps to ensure that the change in the distribution at each step is infinitesimal. This allows the reverse steps $q(x_{t-1}|x_t)$ to be approximated as a Gaussian, which makes the math tractable.

◆ The Learning Goal

The model sees the noisy image (e.g., at $t = 500$) and learns to predict the noise component ϵ that was added. Subtracting this predicted noise moves us on step back toward structure.

The Physics of Diffusion ★ The Reverse Process (Creation)

To generate new data, we must reverse the diffusion. We need to sample from the posterior $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$. However, this distribution is intractable because it depends on the unknown entire dataset.

LEARNED REVERSE TRANSITION

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \Sigma_{\theta}(\mathbf{x}_t, t))$$

KEY VARIABLES

θ

The parameters of our neural network (e.g., U-Net weights).

μ_{θ}

The predicted mean of the previous timestep. This is what the network learns to output.

Σ_{θ}

The variance. Often fixed to $\beta_t I$ or learned separately.



Noisy Input

Neural Network

Predicts μ_{θ} or Noise ε_{θ}



Less Noisy

“Denoising” = Removing Entropy

The Physics of Diffusion ★ Connection to Langevin Dynamics

Sampling from the model is physically equivalent to a particle moving in a potential energy landscape defined by the data distribution.

○ The Score Function

The “force” driving the particle is the gradient of the log-density, $\nabla_{\mathbf{x}} \log p(\mathbf{x})$. It points toward high-density regions (real galaxies).

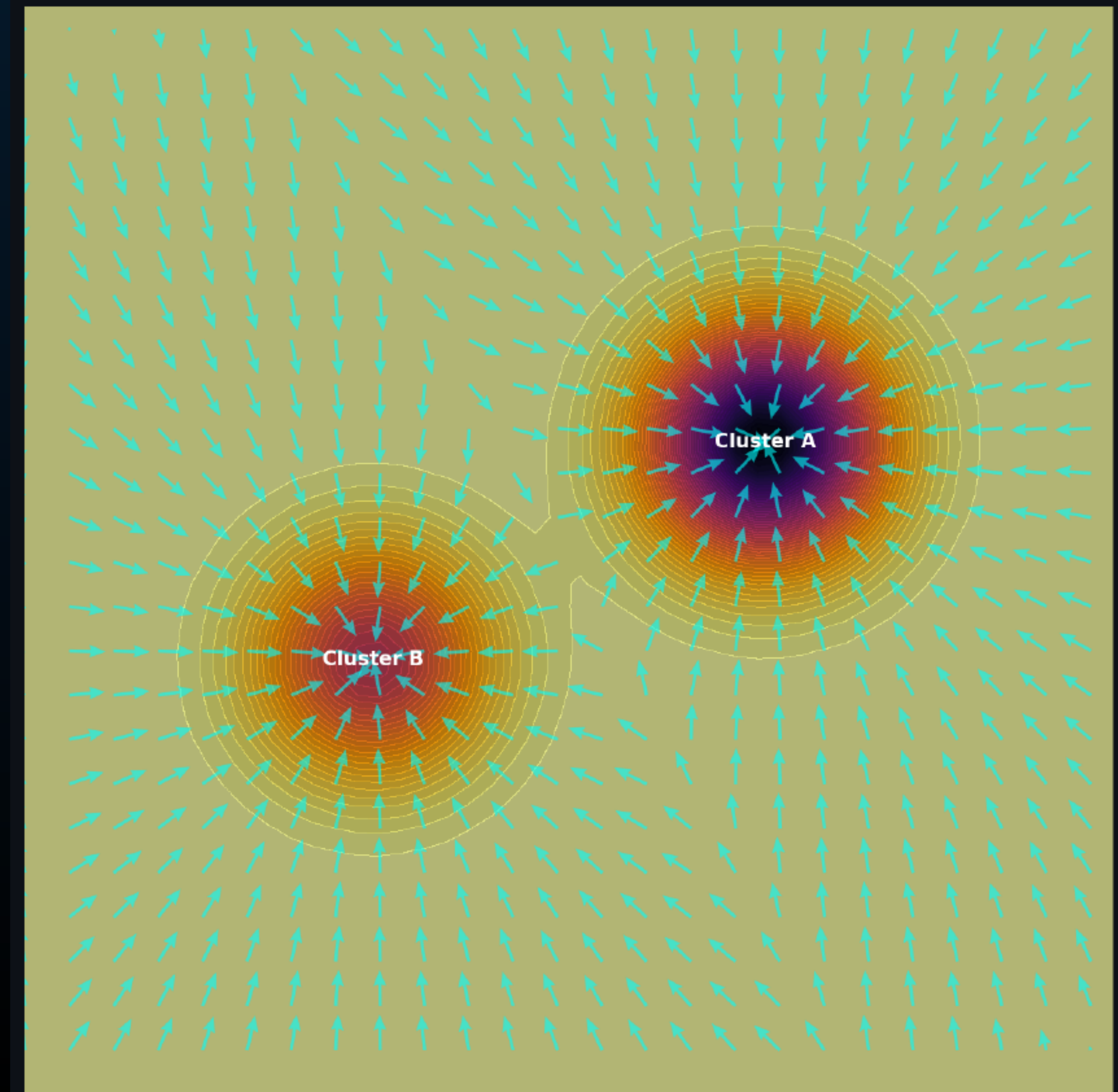
○ Stochastic Exploration

We add noise to prevent getting stuck in local optima, ensuring we sample the full distribution.

DISCRETE LANGEVIN EQUATION

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\delta}{2} \nabla_{\mathbf{x}} \log p(\mathbf{x}_t) + \sqrt{\delta} \epsilon_t$$

$\mathbf{x}$: state δ : step size ϵ : noise



The Physics of Diffusion ★ Score Matching

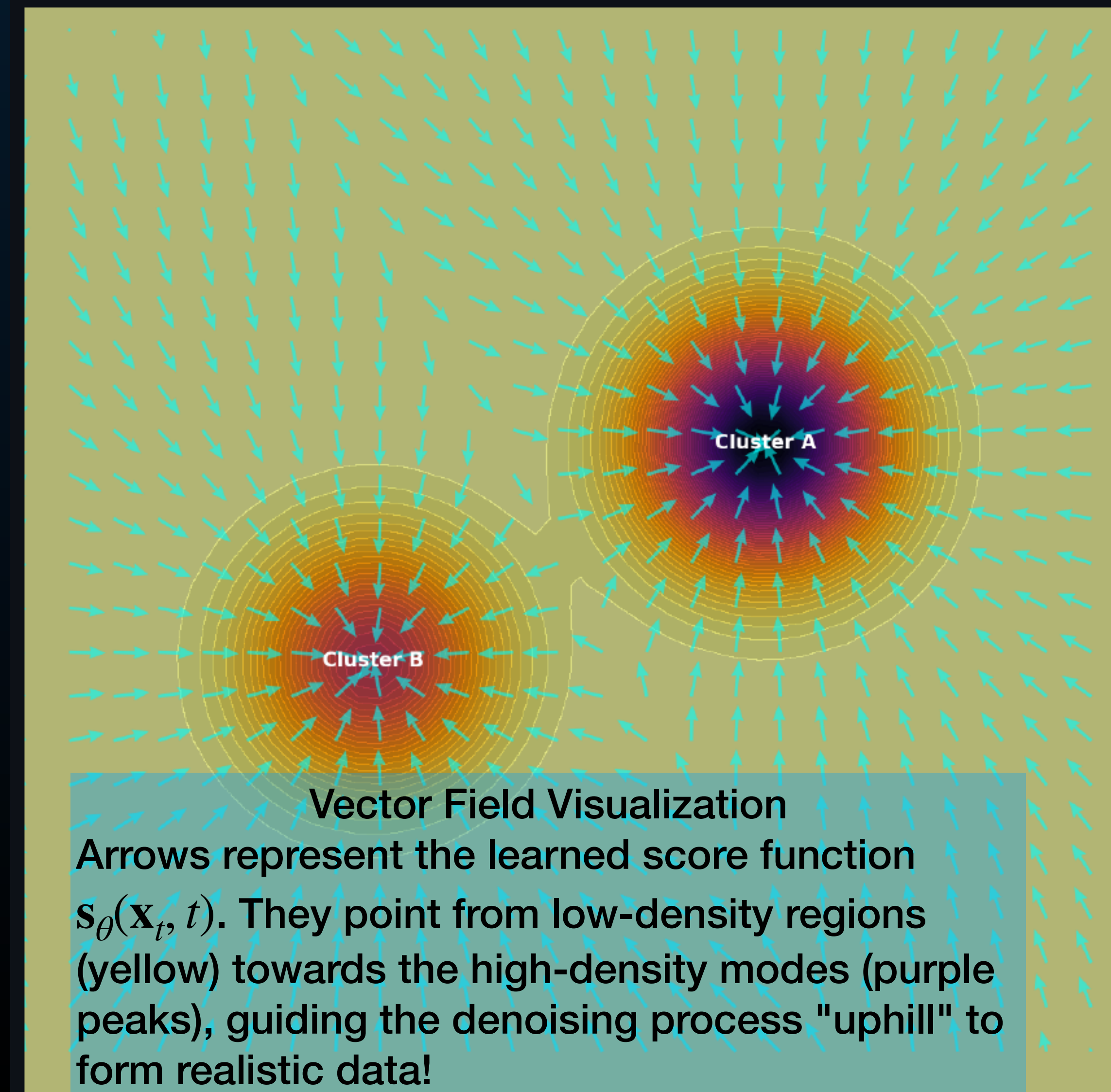
We want to learn the vector field pointing towards high-density regions. This vector field is the Score Function, defined as $\nabla_x \log p(x)$.

DENOISING SCORE MATCHING OBJECTIVE

$$\mathcal{L}_{DSM}(\theta) = \mathbb{E}_{q(\mathbf{x}_0)} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2} \|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0)\|^2 \right]$$

Key Insight: $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0)$ is proportional to the noise $-\varepsilon$.

- 👁 The network $\mathbf{s}_\theta(\mathbf{x}_t, t)$ learns to estimate the gradient of the log-density at any noise level t .
- 👁 By minimizing this Euclidean distance, the model learns “where the data is” relative to the current noisy point.
- 👁 Predicting the added noise ε is equivalent to predicting the score (up to a scaling factor).



The Physics of Diffusion ★ Continuous-Time View (SDEs)

The discrete diffusion process can be generalized to continuous time using **Stochastic Differential Equations (SDEs)**. This connects generative modeling to stochastic calculus.

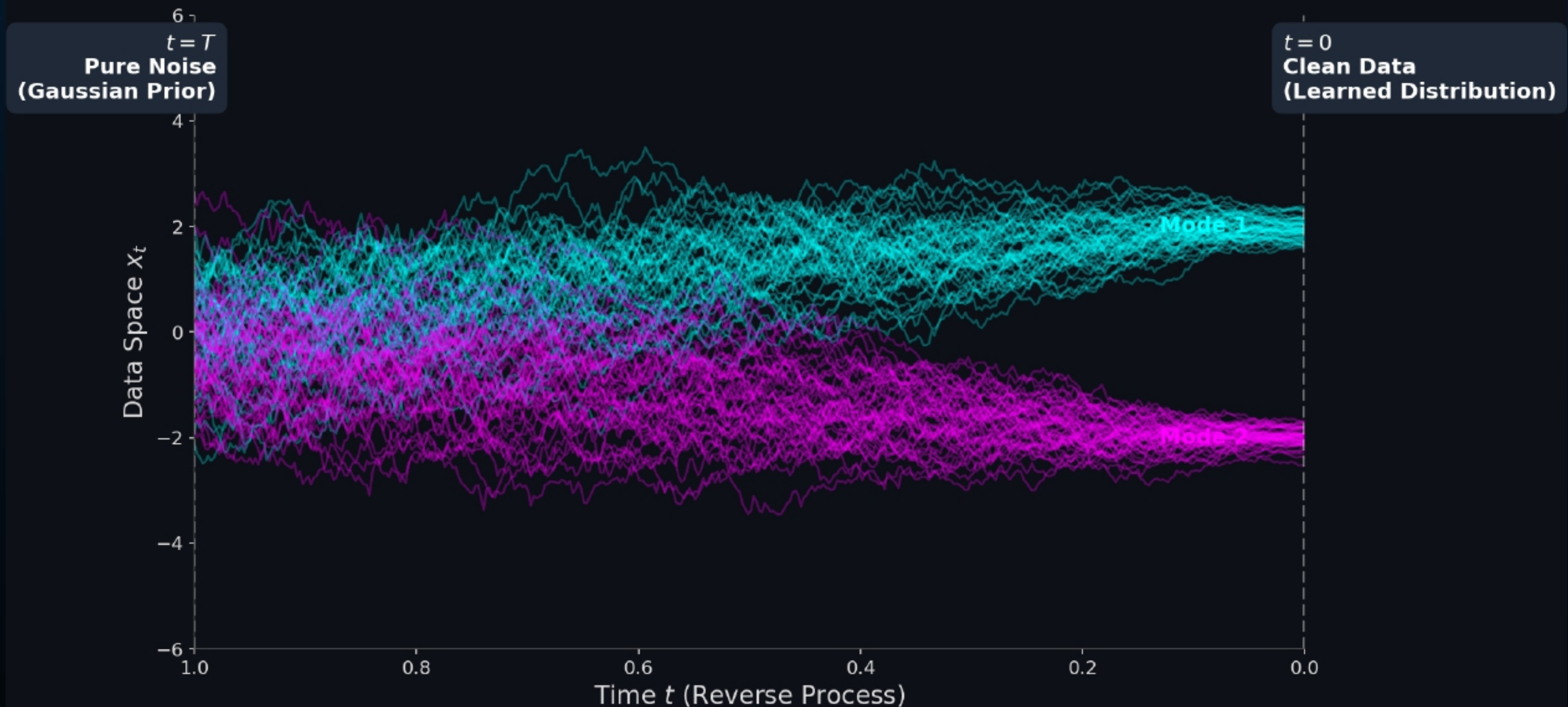
KEY VARIABLES

$\mathbf{f}(\mathbf{x}, t)$ the drift coefficient

$g(t)$ the diffusion coefficient

$\mathbf{W}$ the standard Wiener process (Brownian motion)

Reverse SDE Trajectories: Solving the Inverse Problem



FORWARD SDE (DIFFUSION)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

REVERSE SDE (GENERATIVE)

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + g(t)d\bar{\mathbf{w}}$$

Architecture & Training ★ The Denoiser Learns Noise

A key insight of DDPM is parameterization. Instead of predicting the clean image $\mathbf{x}_0$ directly, the network learns to predict the noise added at timestep t . This simplifies the task to a regression problem.

SIMPLIFIED TRAINING OBJECTIVE (MSE)

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2 \right]$$

Noisy Input $\mathbf{x}_t$

COMPONENTS

ϵ	The ground truth Gaussian noise added to the image (Target).
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ϵ_{θ}	The neural network output (prediction)
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Noisy Image

Time

$\mathbf{x}_t$

t

neural network

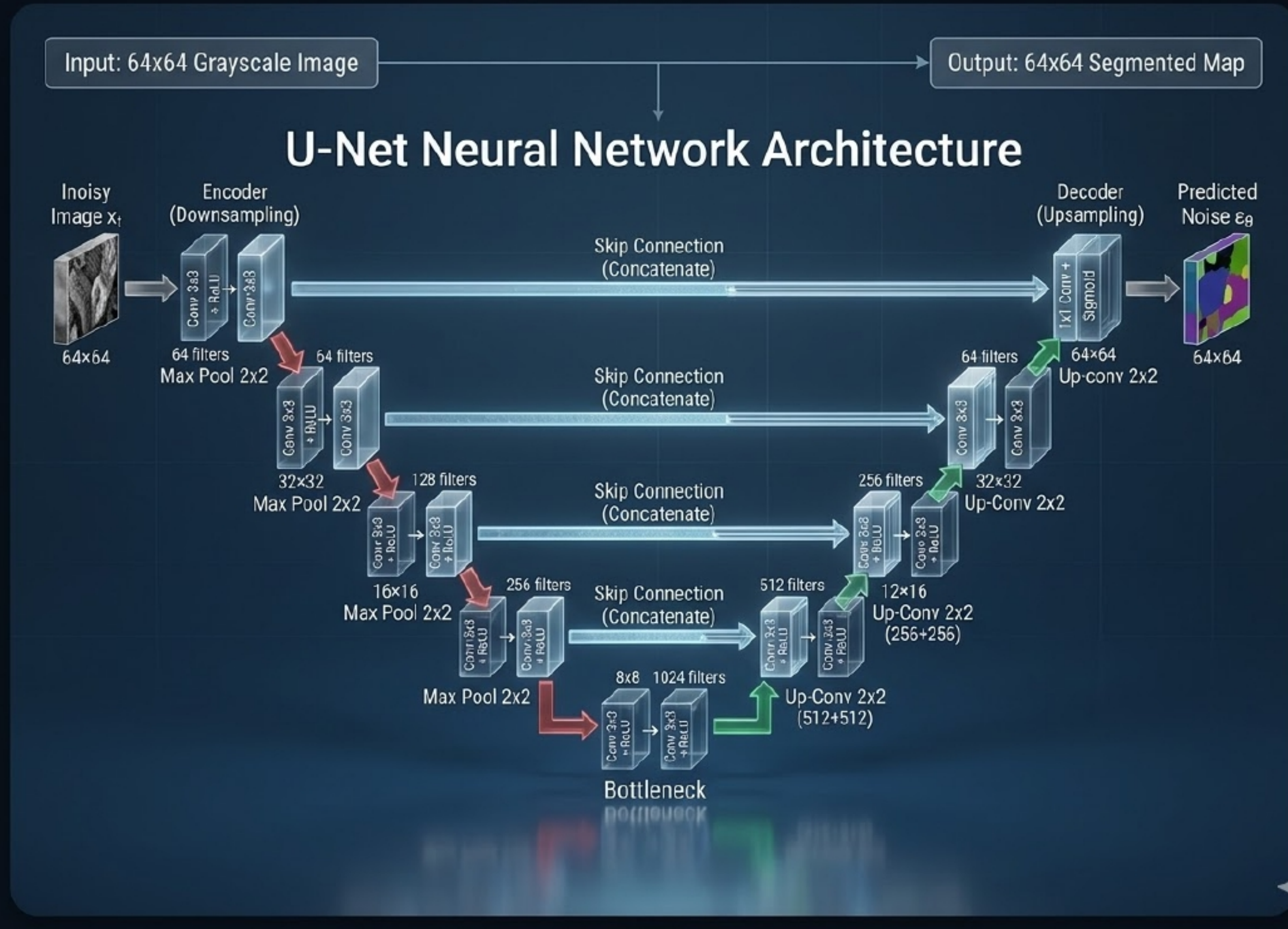
$\epsilon_{\theta}(x_t, t)$

$\epsilon_{\theta}(x_t, t)$

COMPARE
MSE LOSS

ϵ

Architecture & Training ★ The U-Net Backbone



Diffusion models don't use arbitrary networks; they rely on the **U-Net** to process images across multiple scales simultaneously.

Multi-Scale Processing

The Encoder compresses the image to capture global context (morphology), while the Decoder expands it back to pixel space.

Skip Connections

Horizontal connections bypass the bottleneck, preserving high-frequency details like stars and noise textures essential for sharp synthesis.

Input & Output

Unlike classifiers, the U-Net maps $Image(x_t) \rightarrow Noise(\epsilon)$. It preserves spatial dimensions exactly (e.g., 64x64 in, 64x64 out).

Architecture & Training ★ Time Embedding

The U-net shares weights across all time steps. To denoise correctly, it must know the current noise level σ_t . We inject this information via sinusoidal position embedding, adapted from Transformer architectures.

SINUSOIDAL ENCODING FORMULA

$$PE(t, 2i) = \sin\left(\frac{t}{10000^{2i/D}}\right)$$

$$PE(t, 2i + 1) = \cos\left(\frac{t}{10000^{2i/D}}\right)$$

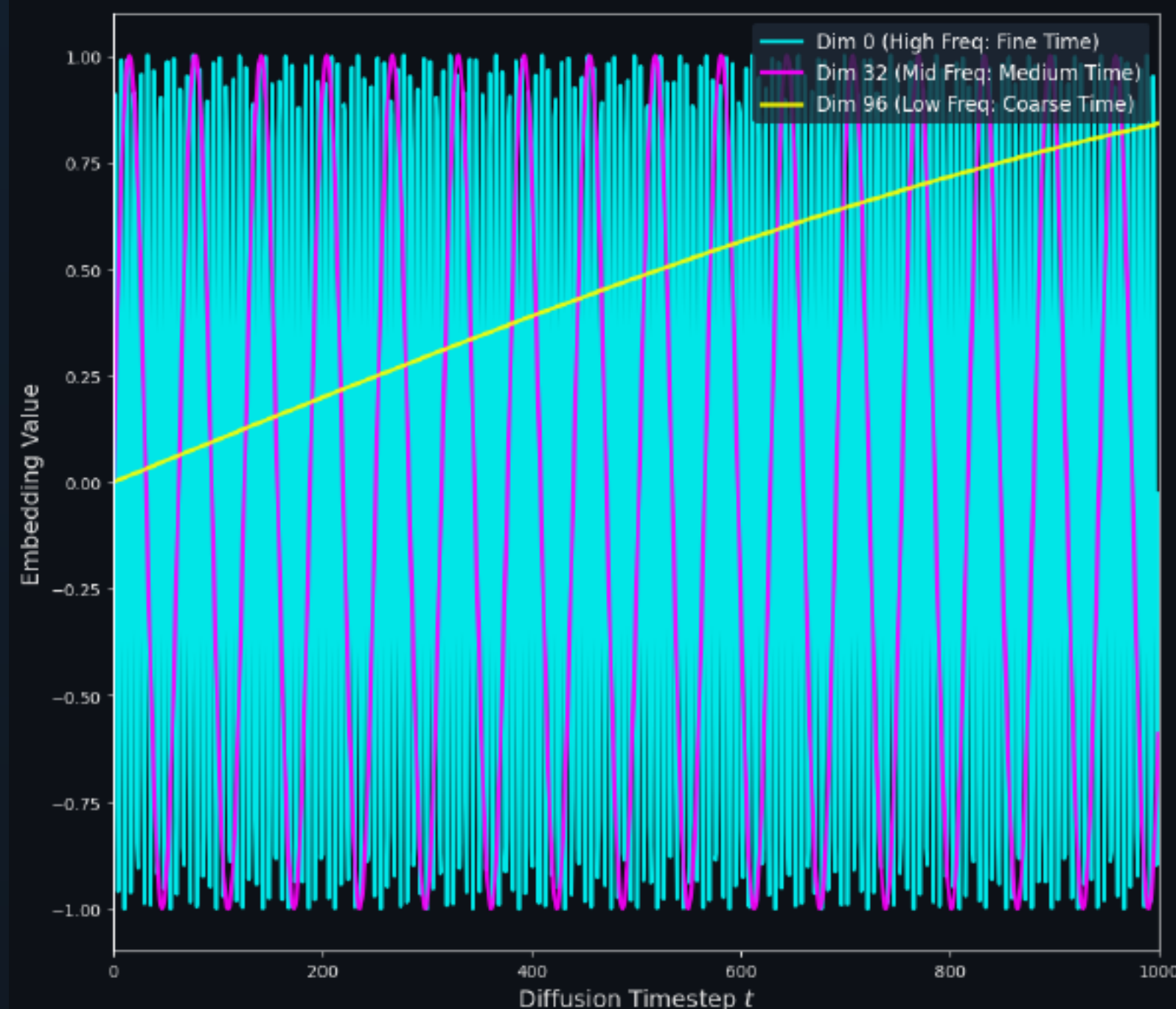
INJECTION MECHANISM

t the current diffusion time step

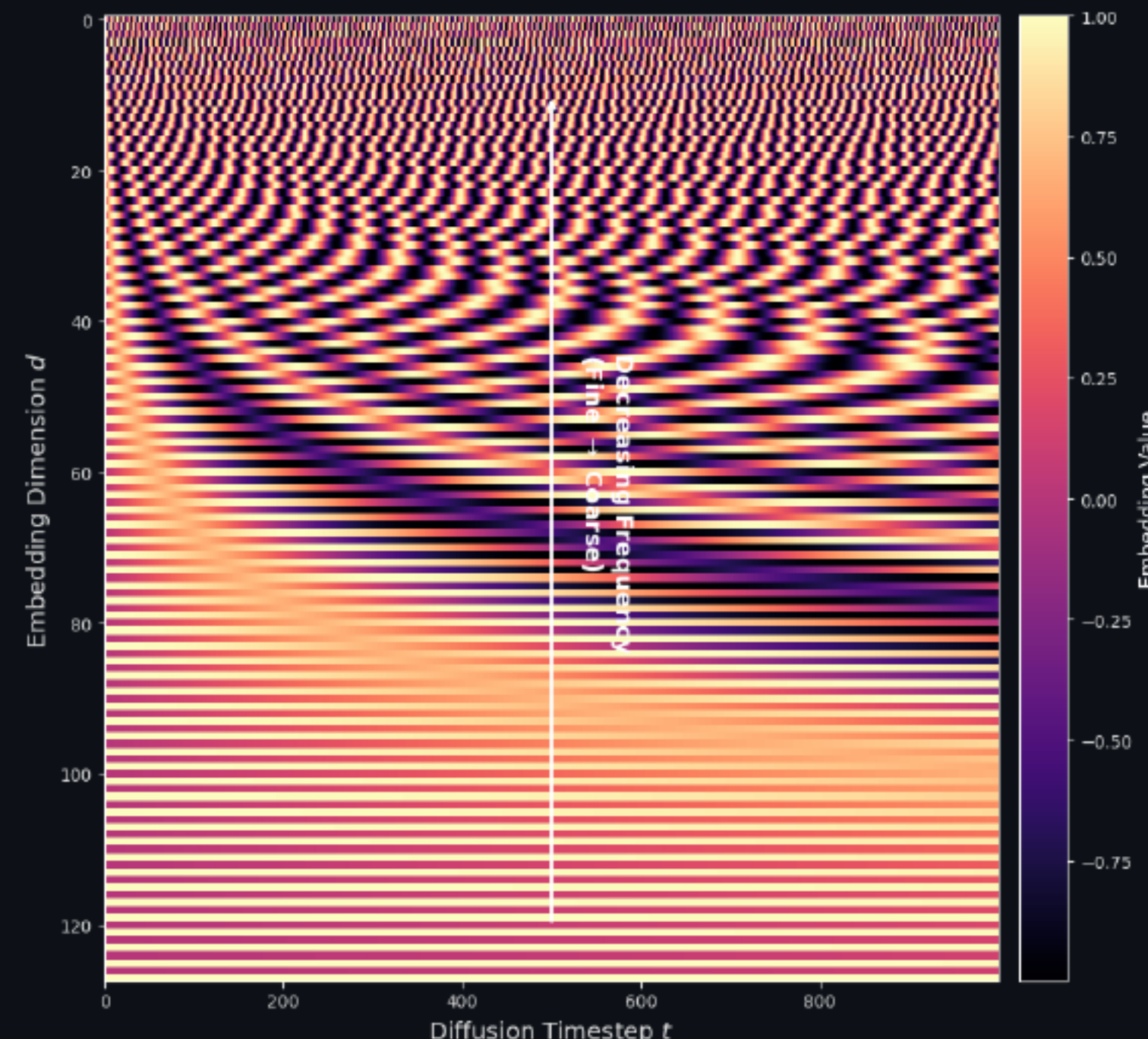
i in $[0, 1, \dots, D/2 - 1]$ is the index of the dimension

D the model's chosen embedding dimension (e.g., 128, 256)

1D View: Tracking Time at Different Scales



2D View: The Full Time Embedding Matrix



Architecture & Training ★ The Computational Bottleneck & Solutions



Why It's Slow

Iterative Refinement Cost

KEY CHALLENGES

- ⌚ **Serial Process:** The Markov chain is inherently serial. Step $t-1$ strictly waits for step t , preventing parallel generation.
- ⌚ **High NFE:** This leads to a huge Number of Function Evaluations (NFE ~ 1000). Generating massive astronomical catalogs (e.g., 10^6 galaxies) becomes computationally intractable.
- ⌚ **Heavy Network Cost:** Every single step requires a full pass through a massive U-Net (often 100M+ parameters) just to predict noise.



How to Accelerate

Samplers & Distillation

ACCELERATION STRATEGIES

- 🚀 **Advanced Samplers (DDIM/PNDM):** Reformulates the process as an ODE, allowing us to safely "skip" steps (e.g., jumping 20 timesteps at once), reducing NFE to ~ 50 with minimal quality loss.
- 🚀 **High-Order Solvers (DPM-Solver):** Employs higher-order Taylor expansions to navigate the curvature of the trajectory, enabling ultra-fast sampling in just 10–20 steps.
- 🚀 **Progressive Distillation:** Trains a smaller "student" model to replicate two steps of a "teacher" model in a single jump, iteratively collapsing the process down to 4–8 steps.

Applications in Astronomy ★ Galaxy Morphology Synthesis

Diffusion models have surpassed GANs in generating **physically realistic** galaxy images that capture the complex, non-Gaussian nature of the Universe.

■ Photorealistic Fidelity

Generated samples exhibit realistic spiral arms, dust lanes, and clumpiness indistinguishable from Hubble or JWST observations, unlike the smooth “blur” of VAEs.

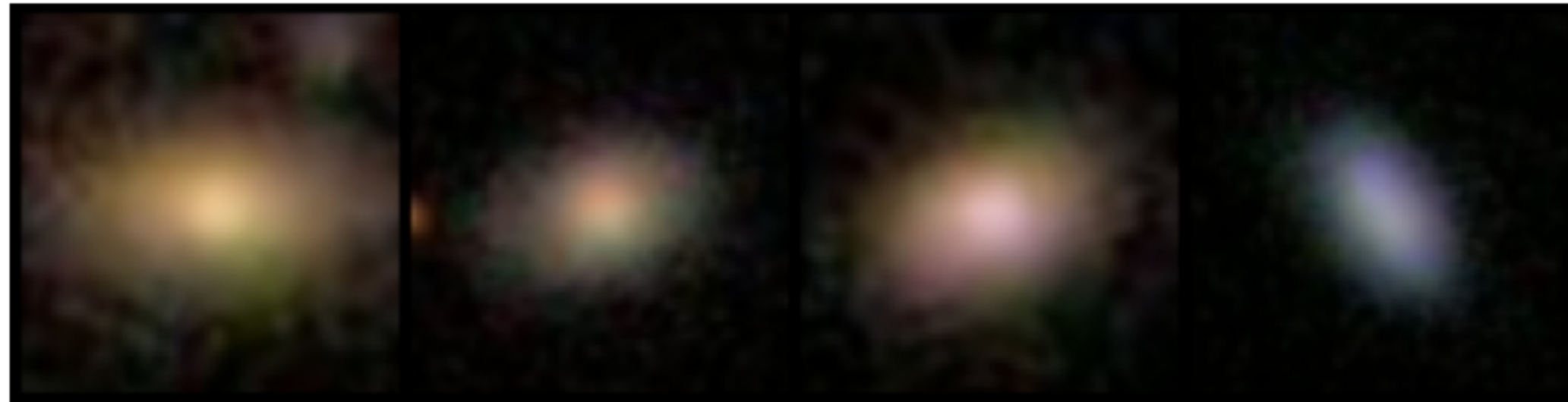
■ Survey Augmentation

We can generate infinite catalogs of “mock” galaxies to stress-test processing pipelines for surveys like LSST and Euclid, correcting for selection biases.

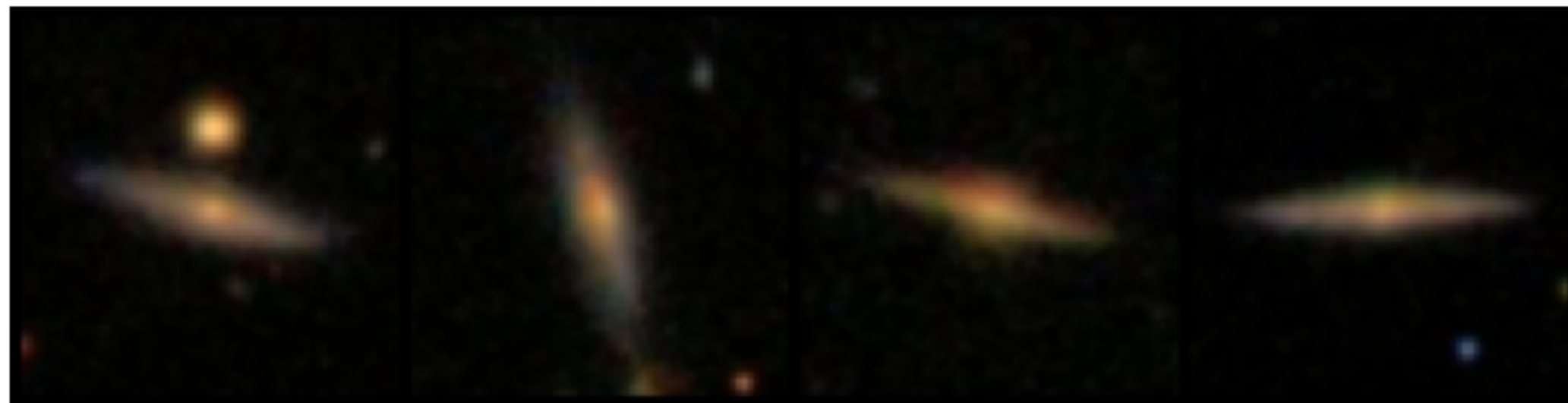
■ Morphological Diversity

The model captures the full diversity of galaxy types—from grand design spirals to irregular mergers—without the “mode collapse” often seen in GANs.

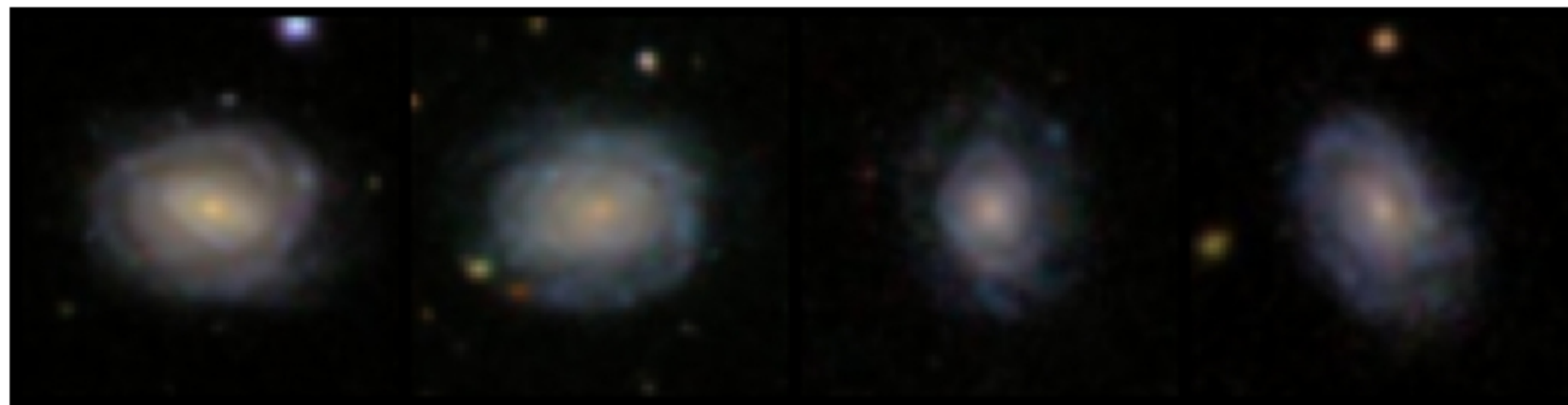
Synthetic Galaxies: Class 2



Synthetic Galaxies: Class 4



Synthetic Galaxies: Class 7



Applications in Astronomy ★ **Conditioning: Asking for Specific Galaxies**

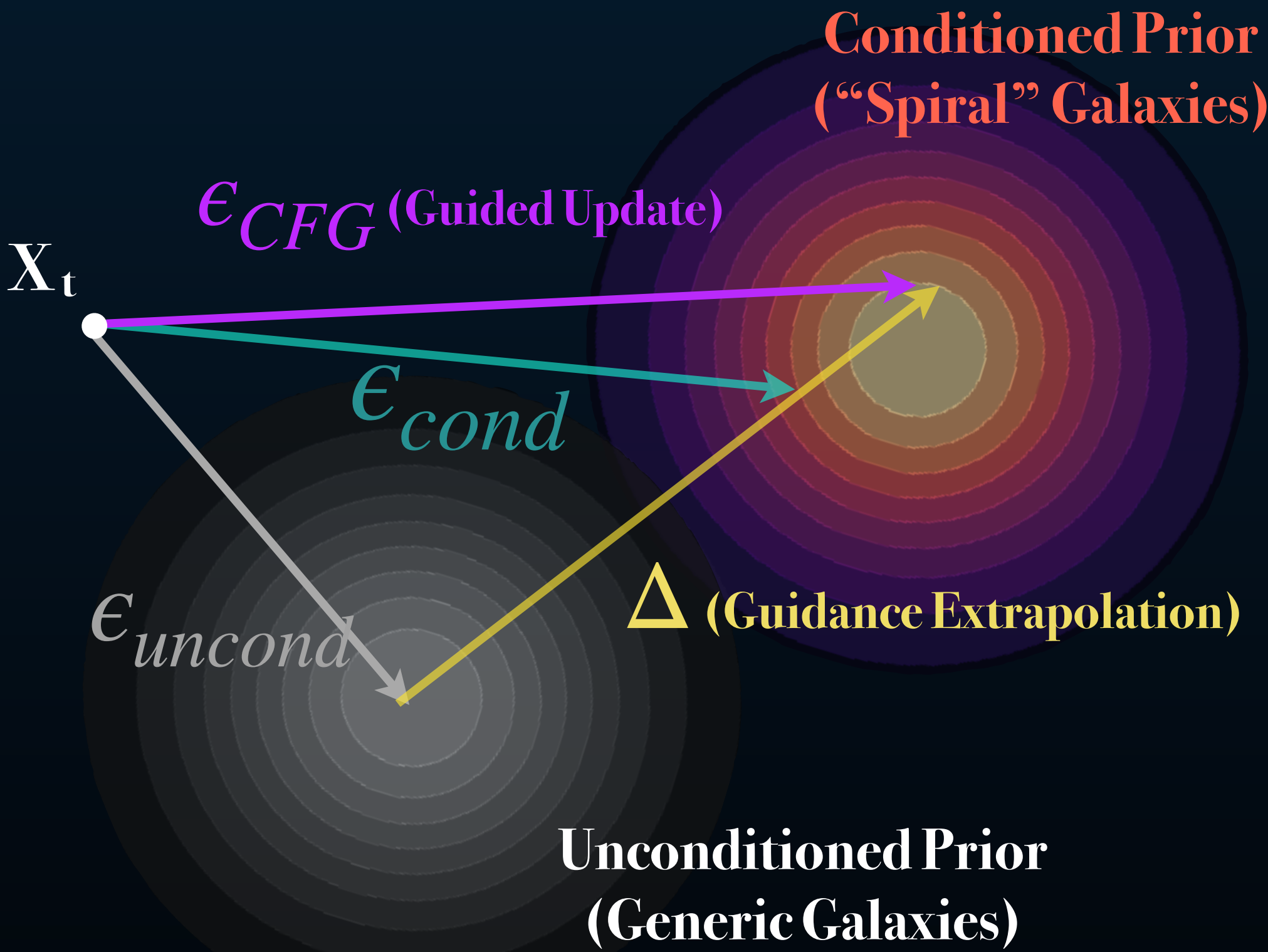
To generate specific objects (e.g., “a spiral at $z=0.5$ ”), we train the denoiser on both conditioned and unconditioned data. During sampling, we trade off diversity for fidelity using **Classifier-Free Guidance**.

GUIDED NOISE PREDICTION

$$\hat{\epsilon}_{\theta}(x_t, c) = \epsilon_{\theta}(x_t, \emptyset) + w \cdot [\epsilon_{\theta}(x_t, c) - \epsilon_{\theta}(x_t, \emptyset)]$$

THE LOGIC OF GUIDANCE

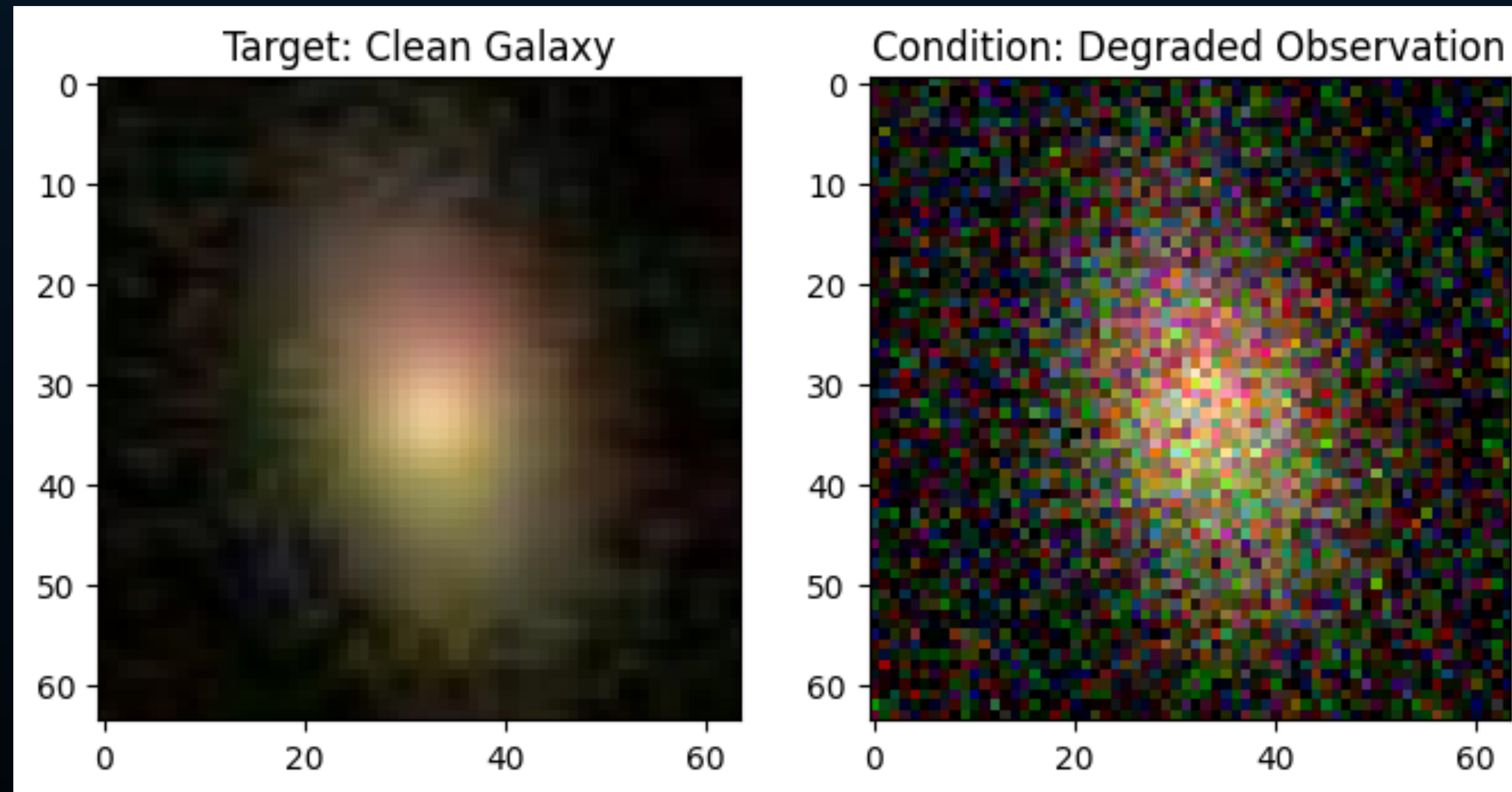
c	The condition (e.g., class label, redshift, text prompt).
$\emptyset$	Null condition.
w	The guidance scale.



$$\epsilon_{CFG} = \epsilon_{uncond} + w \cdot (\epsilon_{cond} - \epsilon_{uncond})$$

By By setting $w > 1$, we extrapolate the difference between the conditioned and unconditioned predictions, pushing the sample deep into the target manifold.

Applications in Astronomy ★ Inverse Problems: Deconvolution



Recovering sharp structure from blurred telescope images is an ill-nosed inverse problem. We use the diffusion model as a **learned prior**.

○ The Observation Model

Telescopes blur the truth x with a Point Spread Function (PSF) H and add noise n .

$$y = Hx + n$$

○ Bayesian Inference

We want the posterior $P(x|y)$. The diffusion model provides the prior $P(x)$, representing “what a galaxy looks like.”

$$P(x|y) \propto P(y|x) \cdot P(x)$$

○ Measurement Consistency

During sampling, we nudge the generated image to ensure it matches the observed data when re-blurred: $\|y - Hx_t\|$.

★ Two Demos



Demo I

Controlled Galaxy Synthesis (Text-to-Image)

Goal

Generate specific galaxies on demand (e.g., edge-on disks, smooth ellipticals) using the Galaxy10 SDSS dataset.

Method

Training a conditional DDPM where morphological class labels guide the reverse diffusion process.

Application

Creating precise synthetic data to correct survey biases and augment training sets for classification models.



Demo II

Astronomical Image Restoration via Image-to-Image DDPMs

Goal

Recover sharp ground-truth structure from degraded observations (PSF blurring, atmospheric seeing, sensor noise).

Method

Training a conditional DDPM where the corrupted image itself acts as the continuous condition, concatenated directly into the U-Net.

Application

Solving ill-posed inverse problems to clean real telescope data without introducing non-physical artifacts.